Project Proposal

Team C4

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1 Team members

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2 Introduction

The proposed project applies to IoT and embedded systems concepts in the domain of automated industrial sorting systems. The system combines machine vision, embedded controllers, sensors, actuators, and wireless communication to automatically identify objects and sort them.

The target application of this project is an automated industrial sorting prototype capable of classifying and separating mechanical fasteners using machine vision and embedded control systems. The system is intended to simulate a mechanical workshop where a lot of small objects are scattered and needs to be sorted in an automated manner.

3 Concept description

The main purpose of the prototype is to demonstrate:

- Distributed embedded systems
- Industrial automation concepts
- Real-time machine vision
- Embedded communication protocols
- Automated object classification and sorting

3.1.1 Devices

- Raspberry Pi 5
- ESP32 Microcontroller
- Raspberry Pi Camera Module (or USB based camera)

3.1.2 Sensors

- IR break-beam sensor
- Rotary encoder
- Camera sensor

3.1.3 Actuators

- Servo motor
- Conveyor DC motor

3.1.4 Communication

- MQTT over Wi-Fi

3.1.5 Software and Tools

- Python
- C/C++
- OpenCV
- TensorFlow Lite
- Arduino IDE
- Mosquitto MQTT Broker

4 Technologies

The proposed system combines embedded Linux computing, lightweight edge AI inference, real-time embedded control, wireless communication, and industrial automation concepts to create an automated machine vision-based sorting system.

A Raspberry Pi 5 will be used for image acquisition, machine vision processing, and object classification using OpenCV and TensorFlow Lite, while an ESP32 microcontroller will handle low-level actuator control, conveyor timing, and servo-based sorting operations. The system will use sensors such as a camera module for image capture, IR sensors for object detection, and a rotary encoder for conveyor position tracking.

Communication between the Raspberry Pi and ESP32 will be implemented using MQTT over Wi-Fi due to its lightweight and low-latency characteristics, making it suitable for IoT and distributed embedded systems.

The software stack will primarily use Python for image processing and MQTT communication, and C/C++ for embedded firmware development and motor control.

5 References

https://github.com/MAliSohail/Team-C4-Advanced-Embedded-Systems-Lab